Assignment 5

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JOIN: Inner / Right+Left / Self

- **Inner Join** is like the AND (∩) logical function, so it will include only where A and B and overlapping, like the center of a Venn Diagram
- **Left/Right Join** will include Left (All of A + A∩B) or Right (All of B + A∩B), and will leave non-matching entries on the 'other' table as NULL
- **Self Join** is used for hierarchical structures where multiple instances of the same data must be loaded. It loads a table to itself so it can be used an an instance for multi-part queries.

1. **List employee names and their department names (Only assigned)**
   - SELECT e.EmployeeName, d.DepartmentName
   - FROM employees AS e
   - INNER JOIN departments AS d ON e.DepartmentID = d.DepartmentID;

```
MariaDB [JoinExamples]> SELECT e.EmployeeName, d.DepartmentName
    -> FROM employees AS e
    -> INNER JOIN departments AS d ON e.DepartmentID = d.DepartmentID;
+--------------+----------------+
| EmployeeName | DepartmentName |
+--------------+----------------+
| Alice        | HR             |
| Bob          | IT             |
| David        | Finance        |
+--------------+----------------+
3 rows in set (0.009 sec)
```

2. List employee names and their department names including those who are not assigned a department.

   - SELECT e.EmployeeName, d.DepartmentName
   - FROM employees AS e
   - LEFT JOIN departments AS d ON e.DepartmentID = d.DepartmentID;

```
MariaDB [JoinExamples]> SELECT e.EmployeeName, d.DepartmentName
    -> FROM employees AS e
    -> LEFT JOIN departments AS d ON e.DepartmentID = d.DepartmentID;
+--------------+----------------+
| EmployeeName | DepartmentName |
+--------------+----------------+
| Alice        | HR             |
| Bob          | IT             |
| Charlie      | NULL           |
| David        | Finance        |
+--------------+----------------+
```

[AT THIS POINT I UPDATED THE EMPLOYEE DATABASE]

3. List all department names and the employees assigned to them. List all departments, even if they have no employees assigned to them.
   - SELECT d.department_name, e.name
   - FROM departments AS d
   - LEFT JOIN employees AS e ON d.id = e.Department_ID;

```
ariaDB [employee]> SELECT d.department_name, e.name
   -> FROM departments AS d
   -> LEFT JOIN employees AS e ON d.id = e.department_id;
------------------+---------+
 department_name  | name    |
------------------+---------+
 HR               | Charlie |
 HR               | Frank   |
 HR               | Paul    |
 HR               | Wendy   |
 Engineering      | Alice   |
 Engineering      | Bob     |
 Engineering      | Jack    |
 Engineering      | Quincy  |
 Engineering      | Xander  |
 Marketing        | Grace   |
 Marketing        | Olivia  |
 Marketing        | Victor  |
 Finance          | David   |
 Finance          | Emma    |
 Finance          | Rachel  |
 Sales            | Hank    |
 Sales            | Ivy     |
 Sales            | Zach    |
 IT               | Karen   |
 IT               | Laura   |
 IT               | Tracy   |
 Customer Support | Mike    |
 Customer Support | Nina    |
 Customer Support | Uma     |
 IT SECURITY      | NULL    |
 QA               | NULL    |
------------------+---------+
```

4. List all employee names and their manager’s names. List only those employees who have a manager.
   - SELECT e.EmployeeName AS "Employee", m.EmployeeName AS "Manager"
   - FROM employees AS e
   - INNER JOIN employees AS m ON e.ManagerID = m.EmployeeID;

```
MariaDB [employee]> SELECT e.name AS "Employee", m.name AS "Manager"
    -> FROM employees AS e
    -> INNER JOIN employees AS m ON e.manager_id = m.id;
+----------+---------+
| Employee | Manager |
+----------+---------+
| Alice    | Jack    |
| Bob      | Jack    |
| David    | Emma    |
| Frank    | Charlie |
| Ivy      | Hank    |
| Karen    | Laura   |
| Mike     | Nina    |
| Olivia   | Grace   |
| Paul     | Charlie |
| Quincy   | Jack    |
| Rachel   | Emma    |
| Tracy    | Laura   |
| Uma      | Nina    |
| Victor   | Grace   |
| Wendy    | Charlie |
| Xander   | Jack    |
| Zach     | Hank    |
+----------+---------+
```

5. List employees and their department names
   - SELECT e.name, d.department_name
   - FROM employees AS e
   - INNER JOIN departments AS d ON e.department_id = d.id;

```
MariaDB [employee]> SELECT e.name, d.department_name
    -> FROM employees AS e
    -> INNER JOIN departments AS d ON e.department_id = d.id;
+---------+------------------+
| name    | department_name  |
+---------+------------------+
| Alice   | Engineering      |
| Bob     | Engineering      |
| Charlie | HR               |
| David   | Finance          |
| Emma    | Finance          |
| Frank   | HR               |
| Grace   | Marketing        |
| Hank    | Sales            |
| Ivy     | Sales            |
| Jack    | Engineering      |
| Karen   | IT               |
| Laura   | IT               |
| Mike    | Customer Support |
| Nina    | Customer Support |
| Olivia  | Marketing        |
| Paul    | HR               |
| Quincy  | Engineering      |
| Rachel  | Finance          |
| Tracy   | IT               |
| Uma     | Customer Support |
| Victor  | Marketing        |
| Wendy   | HR               |
| Xander  | Engineering      |
| Zach    | Sales            |
+---------+------------------+
```

6. List name and salary of employee with highest salary
   - SELECT name, salary
   - FROM employees
   - WHERE salary = (SELECT MAX(salary) FROM employees);

```
MariaDB [employee]> SELECT name, salary
    -> FROM employees
    -> WHERE salary = (SELECT MAX(salary) FROM employees);
+------+----------+
| name | salary   |
+------+----------+
| Emma | 90000.00 |
+------+----------+
```

7. List name, department_name and salary of employee with highest salary
   - SELECT e.name, d.department_name, e.salary

- FROM employees AS e
- INNER JOIN departments AS d ON e.department_id = d.id
- WHERE e.salary = (SELECT MAX(salary) FROM employees);

```
MariaDB [employee]> SELECT e.name, d.department_name, e.salary
    -> FROM employees AS e
    -> INNER JOIN departments AS d ON e.department_id = d.id
    -> WHERE e.salary = (SELECT MAX(salary) FROM employees);
+------+-----------------+----------+
| name | department_name | salary   |
+------+-----------------+----------+
| Emma | Finance         | 90000.00 |
+------+-----------------+----------+
```